Cyprus University of Technology

CSE 226
Algorithms and Complexity

Lecture 18
Maximum Flow (Part 1)
Μέγιστη Ροή (Μέρος 1)

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Contains edited slides from T. H. Cormen, C. E. Leiserson, R. L. Rivest, C. Stein, E. D. Demaine, and M. Christodoulakis

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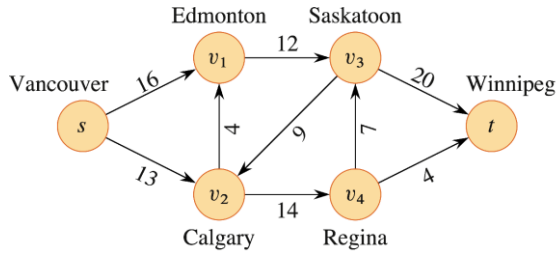
Flow Networks – Δίκτυα Ροής

- A directed graph can be used to model a flow network
  - liquids flowing through pipes
  - parts through assembly lines
  - current through electrical networks
  - …
- The **source** (**πηγή**) s produces the 'material' at some steady rate, and the **sink** (**προορισμός**) t consumes the 'material' at the **same** rate
- *What is the greatest rate at which material can be shipped from the source to the sink?*

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Flow Networks – Definitions

- A **flow network** (**δίκτυο ροής**) $G = (V, E)$ is a directed graph
  - Each edge $(u, v) \in E$ has **capacity** (**χωρητικότητα**) $c(u, v) \geq 0$
  - Two special vertices: **source** (**πηγή**) s and **sink** (**προορισμός**) t
  - Assumption:
    - For any vertex $v \in V$, there is a path $s \rightsquigarrow v \rightsquigarrow t$

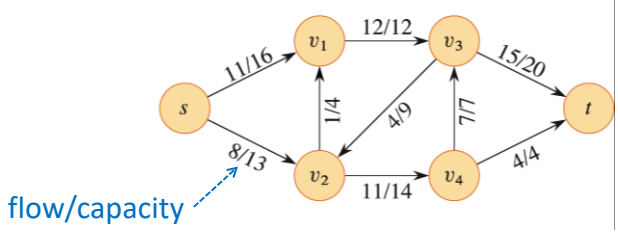


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Flow Networks – Definitions

- A **flow** (**ροή**) in G is a function $f: V \times V \rightarrow \mathbb{R}$ satisfying:
  - **Capacity constraint** (**περιορισμός χωρητικότητας**):
    For all $u, v \in V$, $0 \leq f(u, v) \leq c(u, v)$
  - **Flow conservation** (**διατήρηση ροής**):
    For all $u \in V - \{s, t\}$, $\sum_{v \in V} f(v, u) = \sum_{v \in V} f(u, v)$
    Note: $(u, v) \notin E \Rightarrow f(u, v) = 0$

flow into u flow out of u



flow/capacity

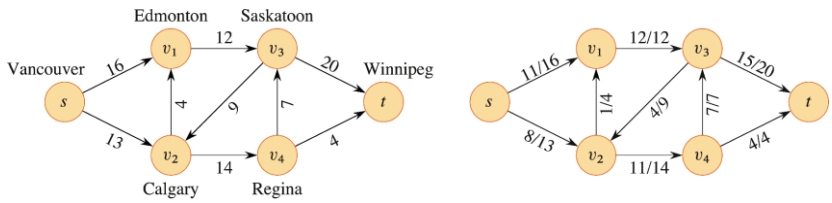
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Value of Flow

- The **value** of a flow (**τιμή** της ροής) f is defined as

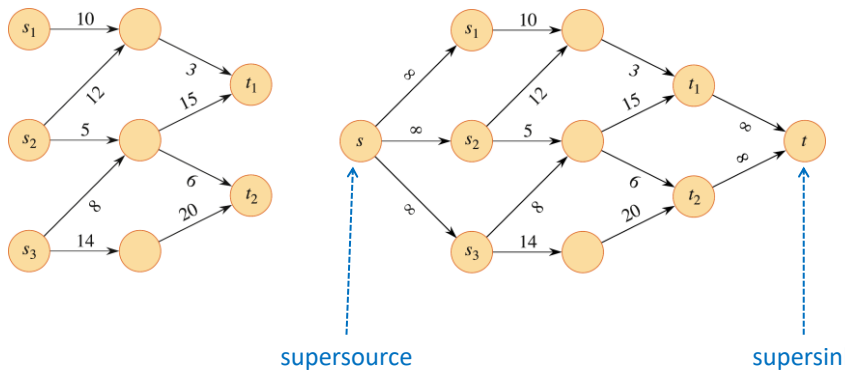
$$|f| = \sum_{v \in V} f(s, v)$$

- In the **maximum-flow problem** (**πρόβλημα της μέγιστης ροής**), we wish to find a flow of maximum value from s to t



Multiple Sources and Sinks

- Fortunately, this is no harder than if we had one source!



The Ford-Fulkerson Method

- "Method" rather than "algorithm"
  - Encompasses several implementations with different running times
- Depends on three important notions
  - **Residual network** (**Υπολειπόμενο δίκτυο**)
  - **Augmenting path** (**Αυξητική διαδρομή**)
  - **Cut** (**Τομή**)

Residual Network

- *Intuition*: Define a graph G_f with edge capacities representing how the flow *can change* in original graph G
  - We may want to *increase* the flow on an edge or we may want to *decrease* the flow on an edge
- **Residual capacity** (**Υπολειπόμενη χωρητικότητα**):

$$c_f(u,v) = \begin{cases} c(u,v) - f(u,v) & \text{if } (u,v) \in E \\ f(v,u) & \text{if } (v,u) \in E \\ 0 & \text{otherwise} \end{cases}$$

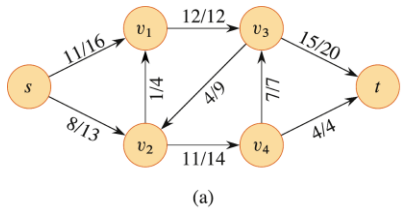
- **Residual network** (**Υπολειπόμενο δίκτυο**):

$$G_f = (V, E_f), \text{ where } E_f = \{(u,v) \in V \times V : c_f(u,v) > 0\}$$

- Observation – edges in E_f are either edges in E or their reversals: $|E_f| \leq 2|E|$

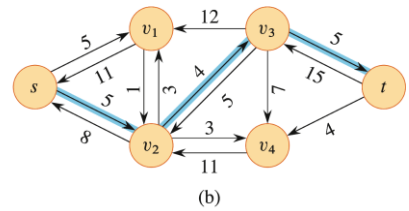
Residual Network – Example

G:

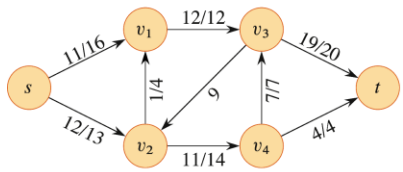


(a)

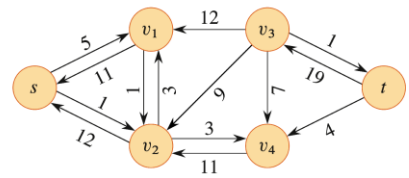
G_f:



(b)



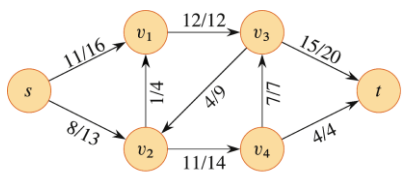
(c)



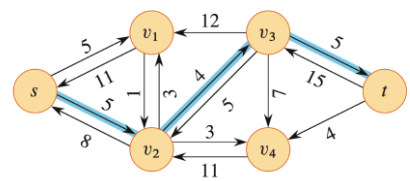
(d)

Augmenting Path

- **Augmenting path** (**Αυξητική διαδρομή**) p is a simple path from s to t in the residual network G_f
- *How much additional flow can we send through an augmenting path*?



Flow Network



Residual Network

Augmenting Path

- **Augmenting path** (**Αυξητική διαδρομή**) p is a simple path from s to t in the residual network G_f
- *How much additional flow can we send through an augmenting path*?
- **Residual capacity** (**Υπολειπόμενη χωρητικότητα**) of p:

$$c_f(p) = \min\{c_f(u,v) : (u,v) \text{ is on } p\}$$

- Doing augmentation: for all (u,v) on p, we just add this $c_f(p)$ to $f(u,v)$ and subtract it from $f(v,u)$
- *Resulting flow is a valid flow with a larger value!*

Flow Augmentation

- **Lemma:** Let $G = (V,E)$ be a flow network with source s and sink t, and let f be a flow in G. Let G_f be the residual network of G induced by f, and let f' be a flow in G_f. Then, the **flow augmentation** (**επαύξηση ροής**) $f \uparrow f'$ is a flow in G with value $|f \uparrow f'| = |f| + |f'|$.
  - $(f \uparrow f')(u,v) =$

$$\begin{cases} f(u,v) + f'(u,v) - f'(v,u) & \text{if } (u,v) \in E \\ 0 & \text{otherwise} \end{cases}$$

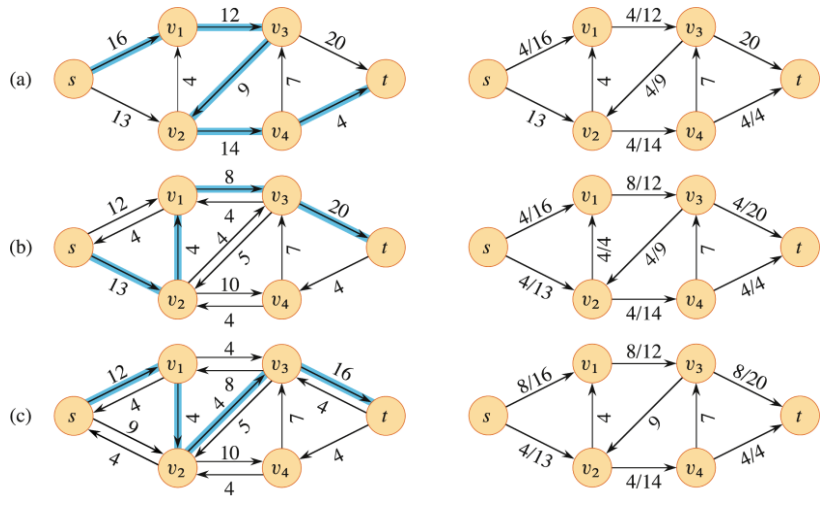
- Intuition:
  - Pushing flow on the reverse edge (v,u) signifies decreasing the flow in (u,v)
  - $f'(u,v) - f'(v,u) \Rightarrow$ keep the net effect of flow on (u,v)

Basic Ford-Fulkerson Algorithm

- **Main Idea:**
  - Start with $f(u, v) = 0$ for all $u, v \in V$
  - At each iteration, increase the flow by finding an augmenting path in a residual network
  - Repeat until no augmenting path can be found

```
FORD-FULKERSON(G, s, t)
1  for each edge (u, v) ∈ G.E
2      (u, v).f = 0
3  while there exists a path p from s to t in the residual network G_f
4      c_f(p) = min{c_f(u, v) : (u, v) is in p}
5      for each edge (u, v) in p
6          if (u, v) ∈ E
7              (u, v).f = (u, v).f + c_f(p)
8          else (v, u).f = (v, u).f − c_f(p)
```


Basic Ford-Fulkerson Algorithm – Example



Basic Ford-Fulkerson Algorithm – Example

